$$\begin{cases} \overline{\nabla U_{l,-}}^2 = 2\sum\limits_{i=2}^{nl/2}(U_i - U_{i-1})^2/nl \\ \overline{\nabla U_{l,+}}^2 = 2\sum\limits_{i=nl/2+1}^{nl}(U_i - U_{i-1})^2/nl \end{cases} \qquad \eta_l = \left(\sum_{k=1}^{l}\overline{\nabla U_{k,-}}^2 + \overline{\nabla U_{k,+}}^2\right)^{\frac{1}{2}} \Big/ \sqrt{2Nl}$$

$$\overline{\nabla U_{l,\pm}}^2 \geq \eta_l$$

$$\eta_{l1} = \sqrt{\frac{\overline{\nabla U_{l1,-}}^2 + \overline{\nabla U_{l1,+}}^2}{2N_b}}$$

$$\overline{\nabla U_{l1,-}}^2 \sim \overline{\nabla U_{l1,+}}^2 \sim \overline{\nabla U_{l1,\pm}}^2$$

$$\overline{\nabla U_{l1,\pm}}^2 \geq \eta_{l1} \quad \longrightarrow \quad \overline{\nabla U_{l1,\pm}}^2 \geq \frac{1}{N_b}$$